CAREER POINT

JEE Main Online Exam 2020

Question with Solutions

3rd September 2020 | Shift-II

PHYSICS

Q.1 To raise the temperature of a certain mass of gas by 50ºC at a constant pressure, 160 calories of heat is required. When the same mass of gas is cooled by 100ºC at constant volume, 240 calories of heat is released. How many degrees of freedom does each molecules of this gas have (assume gas to be ideal) ?

(1) 2 (2) 5 (3) 6 (4) 3

Ans. **[3]**

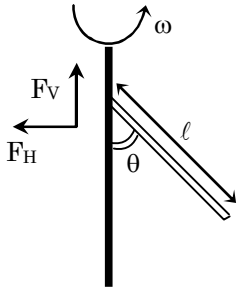
Sol. $nC_p(50) = 160$

$nC_v(100) = 240$

$$\Rightarrow \frac{C_p}{2C_v} = \frac{160}{240} = \frac{\gamma}{2}$$

$$\therefore \gamma = \frac{4}{3} \text{ and } f = \frac{2}{\gamma - 1} = 6$$

Q.2

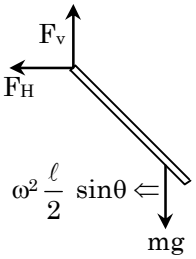


A uniform rod of length 'ℓ' is pivoted at one of its ends on a vertical shaft of negligible radius. When the shaft rotates at angular speed ω the rod makes an angle θ with it (see figure). To find q equate the rate of change of angular momentum (direction going in the paper) $\frac{m\ell^2}{12}\omega^2 \sin\theta\cos\theta$ about the centre of mass (CM)) to the torque provided by the horizontal and vertical forces F_H and F_v about the CM. The value of θ is then such that :

(1) $\cos\theta = \frac{g}{2\ell\omega^2}$ (2) $\cos\theta = \frac{g}{\ell\omega^2}$ (3) $\cos\theta = \frac{2g}{3\ell\omega^2}$ (4) $\cos\theta = \frac{3g}{2\ell\omega^2}$

Ans. **[4]**

Sol.



$F_v = mg$

$F_H = m\omega^2 \frac{\ell}{2} \sin\theta$

$mg \frac{\ell}{2}\sin\theta - m\omega^2 \frac{\ell}{2}\sin\theta \frac{\ell}{2}\cos\theta = \frac{m\ell^2}{12} \omega^2 \sin\theta \cos\theta$

$\cos\theta = \frac{3}{2} \frac{g}{\omega^2\ell}$ ....(ii)

Q.3 The radius R of a nucleus of mass number A can be estimated by the formula $R = (1.3 \times 10^{-15}) A^{1/3}$ m., It follows that the mass density of a nucleus is of the order of :L ($M_{prot.} \simeq M_{neut.} \simeq 1.67 \times 10^{-27}$ kg)

(1) 10^{10} kg m^{-3} (2) 10^{24} kg m^{-3} (3) 10^{17} kg m^{-3} (4) 10^{3} kg m^{-3}

Ans. **[3]**

Sol. $\rho_{nucleus} = \frac{mass}{volume} = \frac{A}{(4/3)\pi r_0^3 A} = \frac{3}{4\pi r_0^3} = 2.3 \times 10^{17}$ kg/m^3

Q.4 The mass density of a planet of radius R varies with the distance r from its centre as $\rho(r) = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$.

Then the gravitational field is maximum at :

(1) $r = \sqrt{\frac{3}{4}} R$ (2) $r = \sqrt{\frac{5}{9}} R$ (3) $r = R$ (4) $r = \frac{1}{\sqrt{3}} R$

Ans. **[2]**

Sol. $E = 4\pi r^2 = \int \rho_0 4\pi r^2 dr$

$\Rightarrow \quad Er^2 = 4\pi G \int_0^r \rho_0 \left(1 - \frac{r^2}{R^2}\right) r^2 dx$

$\Rightarrow \quad E = 4\pi G\rho_0 \left(\frac{r^3}{3} - \frac{r^5}{5R^2}\right)$

$\frac{dE}{dr} = 0 \qquad \therefore \quad r = \sqrt{\frac{5}{9}} R$

Q.5 Amount of solar energy received on the earth's surface per unit area per unit time is defined a solar constant. Dimension of solar constant is -

(1) ML^2T^{-2} (2) $M^2L^0T^{-1}$ (3) MLT^{-2} (4) ML^0T^{-3}

Ans. **[4]**

Sol. $S = \frac{P}{A} = \frac{ML^2T^{-3}}{L^2} = MT^{-3}$

Q.6 A block of mass m attached to a massless spring is performing oscillatory motion of amplitude 'A' on a frictionless horizontal plane. If half of the mass of the block breaks off when it is passing through its equilibrium point, the amplitude of oscillation for the remaining system become fA. The value of f is :

(1) $\sqrt{2}$ (2) 1 (3) $\frac{1}{2}$ (4) $\frac{1}{\sqrt{2}}$

Ans. **[4]**

Sol. At equilibrium position

$V_0 = \omega_1 A = \sqrt{\frac{K}{m}}\, A$ ....(i)

$V = \omega A^1 = \sqrt{\frac{K}{\frac{m}{2}}}\, A^1$ ....(ii)

$\therefore \quad A^1 = \frac{A}{\sqrt{2}}$

Q.7 If a semiconductor photodiode can detect a photon with a maximum wavelength of 400 nm, then its band gap energy is :

Planck's constant $h = 6.63 \times 10^{-34}$ J.s.

Speed of light $c = 3 \times 10^8$ m/s

(1) 3.1 eV (2) 1.1 eV (3) 2.0 eV (4) 1.5 eV

Ans. **[1]**

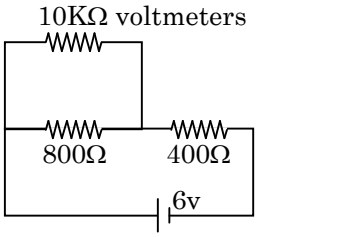
Sol. $\Delta E = \frac{\lambda c}{\lambda e} = 3.1 eV$

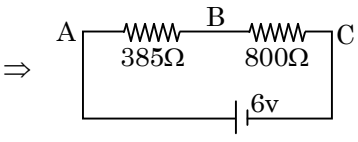
Q.8 Two resistors 400Ω and 800Ω are connected in series across a 6 V battery. The potential difference measured by a voltmeter of 10 kΩ across 400Ω resistor is close to :

(1) 2.05 V (2) 1.95 V (3) 2 V (4) 1.8 V

Ans. **[2]**

Sol.





So the potential difference in voltmeter across the points A and B is $\frac{6}{1185} \times 385 = 1.949$ V

Q.9 A metallic sphere cools from 50ºC to 40ºC in 300 s. If atmospheric temperature around is 20ºC, then sphere's temperature after the next 5 minutes will be close to :

(1) 33ºC (2) 31ºC (3) 35ºC (4) 28ºC

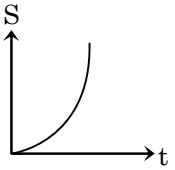
Ans. **[1]**

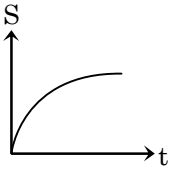
Sol. $$\frac{50-40}{300} = \beta\left(\frac{50+40}{2} - 20\right)$$

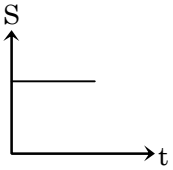
$$\frac{40-T}{300} = \beta\left(\frac{40+T}{2} - 20\right)$$

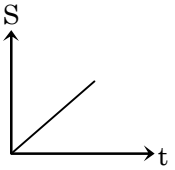
$$\therefore \quad T = \frac{100}{3}$$

Q.10 A particle is moving unidirectionally on a horizontal plane under the action of a constant power supplying energy source. The displacement (s) – time (t) graph that describes the motion of the particle is (graphs are drawn schematically and are not to scale) :

(1) 

(2) 

(3) 

(4) 

Ans. **[1]**

Sol. $$\frac{dK}{dE} = P = \cos t \Rightarrow K = Pt = \frac{1}{2}mV^2$$

$$\therefore \quad V = \sqrt{\frac{2Pt}{m}} = \frac{ds}{dt} \qquad \therefore \quad S = \sqrt{\frac{2P}{m}}\ \frac{2}{3}t^{3/2}$$

Q.11 A calorimeter of water equivalent 20 g contains 180 g of water at 25ºC. 'm' grams of steam at 100ºC is mixed is 31ºC. The value of 'm' is close to (Latent heat of water = 540 cal g^{-1}, specific heat of water = 1 cal $g^{-1}\,{}^{o}C^{-1}$).

(1) 2 (2) 2.6 (3) 4 (4) 3.2

Ans. **[1]**

Sol.

Cal	H_2O	Sterm
20gm	180gm	m
25°C	25°C	100°C

$200 \times 1 \times (31 - 25)$

$= m \times 540 + m \times 1 \times (100 - 31)$

Q.12 Two sources of light emit X-rays of wavelength 1 nm and visible light of wavelength 500 nm, respectively. Both the sources emit light of the same power 200 W. The ratio of the number density of photons of X-rays to the number density of photons of the visible light of the given wavelengths is :

(1) $\frac{1}{500}$ (2) 500 (3) $\frac{1}{250}$ (4) 250

Ans. **[1]**

Sol. $$P = \frac{nhc}{\lambda t}$$

$$\therefore \quad \frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2} = \frac{1}{5}$$

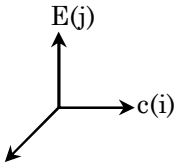
Q.13 The electric field of a plane electromagnetic wave propagating, along the x direction is vacuum is $\vec{E} = E_0 \hat{j} \cos(wt - kx)$. The magnetic field $\vec{B}$, at the moment t = 0 is:

(1) $\vec{B} = \frac{E_0}{\sqrt{\mu_0 \in_0}} \cos(kx)\hat{j}$ (2) $\vec{B} = E_0\sqrt{\mu_0 \in_0} \cos(kx)\hat{j}$

(3) $\vec{B} = \frac{E_0}{\sqrt{\mu_0 \in_0}} \cos(kx)\hat{k}$ (4) $\vec{B} = E_0\sqrt{\mu_0 \in_0} \cos(kx)\hat{k}$

Ans. **[4]**

Sol.



$\therefore \quad \vec{B} \ (\hat{k})$

$\Rightarrow \quad \vec{B} = B_0 \cos(wt - kx)\hat{k}$

Now put t = 0.

Q.14 Hydrogen ion and singly ionized helium atom are accelerated, from rest, through the same potential differentiation. the ratio of final speeds of hydrogen and helium ions is close to -

(1) 1 : 2 (2) 10 : 7 (3) 5 : 7 (4) 2 : 1

Ans. **[4]**

Sol. $q\Delta V = \frac{1}{2}mV^2 \quad \Rightarrow \quad v = \sqrt{\frac{2q\Delta V}{m}}$

$\therefore \quad \frac{V_1}{V_2} = \sqrt{\frac{e}{m}\frac{4m}{e}} = 2$

Q.15 Two light waves having the same wavelength λ in vacuum are in phase initially, then the first wave travels a path L_1 through a medium of refractive index n_1 while the second wave travels a path of length L_2 through a medium of refractive index n_2. After this the phase difference between the two waves to -

(1) $\frac{2\pi}{\lambda}(n_1L_1 - n_1L_2)$ (2) $\frac{2\pi}{\lambda}\left(\frac{L_1}{n_1} - \frac{L_2}{n_2}\right)$ (3) $\frac{2\pi}{\lambda}(n_1L_1 - n_2L_2)$ (4) $\frac{2\pi}{\lambda}\left(\frac{L_2}{n_1} - \frac{L_1}{n_2}\right)$

Ans. **[3]**

Sol. $\Delta p = n_1 L_1 - n_2 L_2$

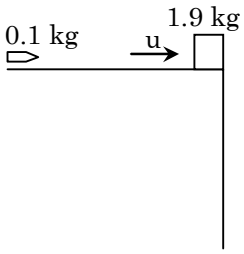
$\Delta\phi = \frac{2\pi}{\lambda}\Delta p$

Q.16 A block of mass 1.9 kg is at rest at the edge of a table, of height 1m. A bullet of mass 0.1 kg collides with the block and sticks to it. If the velocity of the bullet is 20 m/s in the horizontal direction just before the collision then the kinetic energy just before the combined system strikes the floor, is [Take g = 10 m/s^2. Assume there is no rotational motion and loss of energy after the collision is negligible.]

(1) 21 J (2) 23 J (3) 20 J (4) 19 J

Ans. **[1]**

Sol.



$p_i = p_f \Rightarrow 0.1 \times 20 = 2v$

$\therefore \quad v = 1 \text{ m/s}$

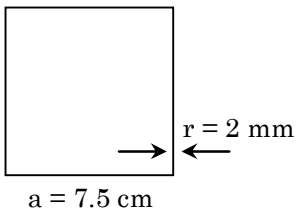
$KE_f = mgh + \frac{1}{2} mv^2 = 213$

Q.17 A uniform magnetic field B exists in a direction perpendicular to the plane of a square loop made of a metal wire. The wire has a diameter of 4 mm and a total length of 30 cm. The magnetic field changes with time at a steady rate $dB/dt = 0.032\ Ts^{-1}$. The induced current in the loop is close to (Resistivity of the metal wire is $1.23 \times 10^{-8}\ \Omega m$)

(1) 0.34 A (2) 0.53 A (3) 0.61 A (4) 0.43 A

Ans. **[3]**

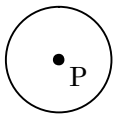
Sol.



$q_i = \frac{d(Ba^2)}{dt} = a^2 \frac{dB}{dt}$

$i = \frac{q}{R} = \frac{a^2 dB/dt}{\frac{\rho(40)}{\pi r^2}}$

Q.18 A perfectly diamagnetic sphere has a small spherical cavity at its centre, which is filled with a paramagnetic substance. The whole system is placed in a uniform magnetic field $\vec{B}$. Then the field inside the paramagnetic substance is :



(1) Zero (2) $\vec{B}$

(3) much large than $|\vec{B}|$ and parallel to $\vec{B}$ (4) much large than $|\vec{B}|$ but opposite to $\vec{B}$

Ans. **[1]**

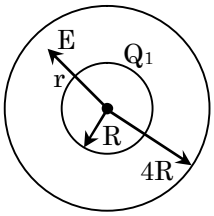
Sol. A perfect diamagnetic substance will completely expel the magnetic field. Therefore, there will be no magnetic field inside the cavity of sphere. Hence the paramagnetic substance kept inside the cavity will experience no force.

Q.19 Concentric metallic hollow spheres radii R and 4R hold charges Q_1 and Q_2 respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference V(R) – V(4R) is :

(1) $\frac{3Q_1}{16\pi\varepsilon_0 R}$ (2) $\frac{3Q_1}{4\pi\varepsilon_0 R}$ (3) $\frac{Q_2}{4\pi\varepsilon_0 R}$ (4) $\frac{3Q_2}{4\pi\varepsilon_0 R}$

Ans. **[1]**

Sol.



$$E = \frac{KQ_1}{r^2}$$

$$\Delta V = \int_R^{4R} E\,dr = \frac{3KQ_1}{4R}$$

Q.20 Which of the following will NOT be observed when a multimeter (operating in resistance measuring mode) probes connected across a component, are just reversed?

(1) Multimeter shows an equal deflection in both cases i.e. before and after reversing the probes if the chosen component is resistor.

(2) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is metal wire.

(3) Multimeter shows a deflection, accompanied by a splash of light out of connected component in one direction and NO deflection on reversing the probes if the chosen component is LED.

(4) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor.

Ans. **[4]**

Sol. (1) Multimeter shows deflection when it connects with capacitor.

(2) If we assume that LED has negligiable resistance then multimeter shows no deflection for the forward blas but when it connects in reverse direction, it break down occurs so splash of light out.

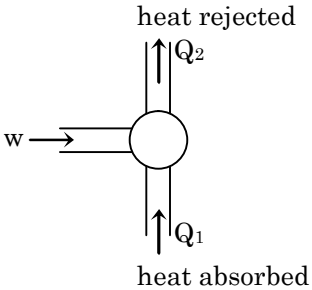
(3) The resistance of metal wire may be taken zero, so no deflection in multimeter.

(4) No matter, how we connect the resistance across multimeter. It shows same deflection.

Q.21 If minimum possible work is done by a refrigerator in converting 100 grams of water at 0ºC to ice, how much heat (in calories) is released to the surroundings at temperature 27ºC (Latent heat of ice = 80 Cal/gram) to the nearest integer?

Ans. **[8791]**

Sol.



$$w + Q_1 = Q_2$$

$$w = Q_2 - Q_1$$

$$\text{C.O.P.} = \frac{Q_1}{w}\ \frac{Q_1}{Q_2 - Q_1} = \frac{273}{300-273} = \frac{Q_1}{W}$$

$$w = \frac{27}{273} \times 80 \times 100 \times 4.2$$

$$Q_2 = w + \theta_1$$

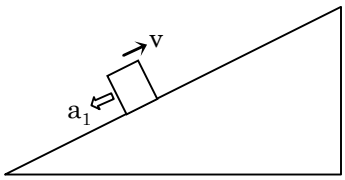
$$Q_2 = \frac{27}{273} \times 80 \times 100 \times 4.2 + 80 \times 100 \times 4.2$$

$$Q_2 = \frac{300}{273} \times 80 \times 100 = 8791.2 \text{ cal}$$

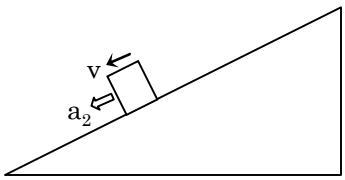
Q.22 A block starts moving up an inclined plane of inclination 30º with an initial velocity of v_0. It comes back to its initial position with velocity $\frac{v_0}{2}$. The value of the coefficient of kinetic friction between the block and the inclined plane is close to $\frac{I}{1000}$. The nearest integer to I is ..... .

Ans. **[346]**

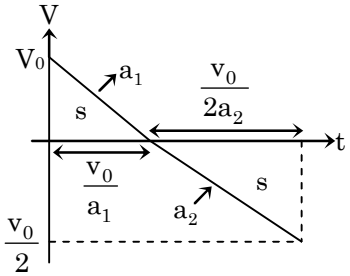
Sol.



$a_1 = g(\sin\theta + \mu\cos\theta)$



$a_2 = g(\sin\theta + \mu\cos\theta)$



$$\therefore \quad \frac{1}{2}\, v_0\, \frac{v_0}{a_1} = \frac{1}{2}\left(\frac{v_0}{2}\right)\left(\frac{v_0}{2a_1}\right)$$

$$\Rightarrow \quad 3\sin\theta = 5\,\mu\cos\theta$$

$$\therefore \quad \mu = \sqrt{3}/5$$

Q.23 A glavanometer coil has 500 turns and each turn has an average area of 3×10^{-4} m^2. If a torque of 1.5 Nm is required to keep this coil parallel to a magnetic field when a current of 0.5 A is flowing through it, the strength of the field (in T) is ..... .

Ans. **[20]**

Sol. $\vec{\tau} = \vec{m} \times \vec{B}$

$\tau = NI \times A \times B$

$105 = 500 \times 3 \times 10^{-4} \times \frac{1}{2} \times B$

$B = 20$

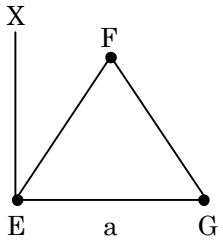
Q.24 When an object is kept at a distance of 30 cm from a concave mirror, the image is formed at a distance of 10 cm from the mirror. If the object is moved with a speed of 9 cms^{-1}, the speed (in cms^{-1}) with which image moves at that instant is ..... .

Ans. **[1]**

Sol. $\left|\left(\frac{dv}{dt}\right)\right| = \left|\frac{v^2}{4^2}\right| \left|\frac{du}{dt}\right|$

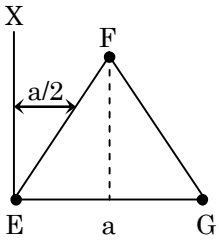
$= \left(\frac{10}{30}\right) 2 \times 9 = 1 m/s$

Q.25 An massless equilateral triangle EFG of side 'a' (As shown in figure) has three particles of mass m situated at its vertices. The moment of inertia of the system about the line EX perpendicular to EG in the plane of EFG is $\frac{N}{20}$ ma^2 where N is an integer. The value of N is .....



Ans. **[25]**

Sol.



$I = 0 + m\left(\frac{a}{2}\right)^2 + ma^2$

$= \frac{5}{4} ma^2$

CAREER POINT

JEE Main Online Exam 2020

Question with Solutions

3rd September 2020 | Shift-II

CHEMISTRY

Q.1 The incorrect statement(s) among (a) – (d) regarding acid ram is (are) :

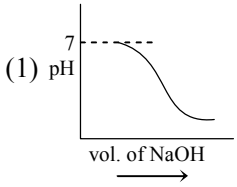
(a) It can corrode water pipes
(b) It can damage structures made up of stone
(c) It cannot cause respiratory ailments in animals
(d) It is not harmful for trees

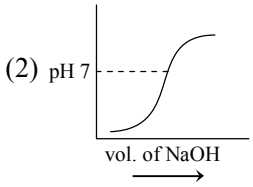
(1) (c) only (2) (c) and (d) (3) (a), (b) and (d) (4) (a), (c) and (d)

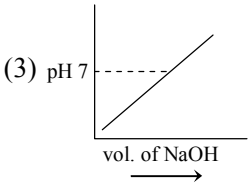
Ans. **[3]**

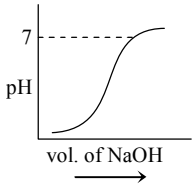
Sol. (1) Acid rain corrodes water pipes resulting in the leaching of heavy of heavy metals such as iron, lead and copper into the drinking water.
(2) Acid rain damages buildings and other structures made of stone or metal.
(3) It causes respiratory aliments in human beings and animals.
(4) It is harmful for agriculture, trees and plants as it wasshes down the nutrients needed for its growth.

Q.2 100 mL of 0.1 M HCl is taken in a beaker and to it 100 mL of 0.1 M NaOH is added in steps 2 mL and the pH is continuously measured. Which of the following graphs correctly depicts the change in pH?

(1) 

(2) 

(3) 

(4) 

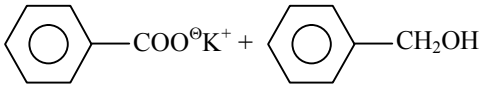
Ans. **[2]**

Sol. Steep rise in pH around the equivalence point for titration of strong acid with strong base.

Q.3 The compound A in the following reactions is :

$$A \xrightarrow[(ii)\ Conc.,\ H_2SO_4/\Delta]{(i)\ CH_3MgBr/H_2O}$$

$$B \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_2} C + D$$

$$C \xrightarrow[(ii)\ \Delta]{(i)\ Conc.\ KOH} \text{  } + \text{}$$

$$D \xrightarrow[\Delta]{Ba(OH)_2} H_3C-\overset{CH_3}{\overset{|}{C}}=CH-\overset{O}{\overset{||}{C}}-CH_3$$

(1) $C_6H_5-\overset{O}{\overset{||}{C}}-CH\begin{matrix}CH_3\\CH_3\end{matrix}$

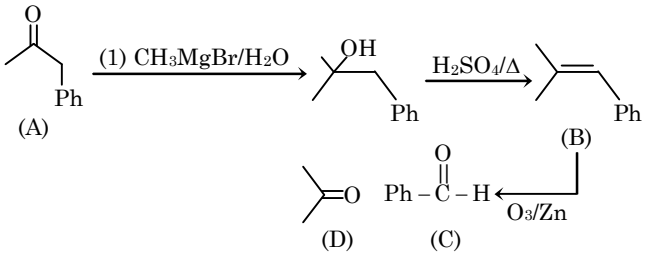
(2) $C_6H_5-CH_2-\overset{O}{\overset{||}{C}}-CH_3$

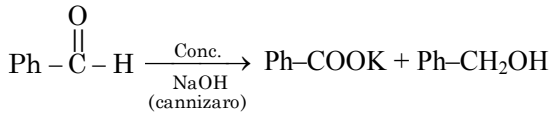
(3) $C_6H_5-\overset{O}{\overset{||}{C}}-CH_2CH_3$

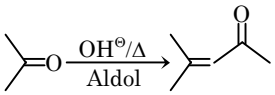
(4) $C_6H_5-\overset{O}{\overset{||}{C}}-CH_3$

Ans. **[2]**

Sol.







Q.4 Consider the hypothetical situation where the azimuthal quantum number, ℓ takes values 0, 1, 2, ... n + 1. where n is the principal quantum number. Then, the element with atomic number.

(1) 6 has a 2p-valence subshell
(2) 8 is the first noble gas
(3) 13 has a half-filled valence subshell
(4) 9 is the first alkali metal

Ans. **[3]**

Sol. $\ell = 0$ to $(n + 1)$

n = 1
l = 0, 1, 2

n = 2
l = 0, 1, 2, 3

$(n + l) \Rightarrow \frac{1s}{1}\ \frac{1p}{2}\ \frac{1d}{3}$ $\frac{2s}{2}\ \frac{2p}{3}\ \frac{2d}{4}\ \frac{2f}{5}$

n = 3

$\ell = 0, 1, 2, 3, 4$

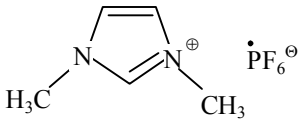
$\frac{3s}{3}\ \frac{3p}{4}\ \frac{3d}{5}\ \frac{3f}{6}\ \frac{3g}{7}$

Now, in order to wire electronic configuration, we need to apply $(n + \ell)$ rule

Energy order 1 : $1s < 1p < 2s < 1\ d < 2p < 3s < 2d$ ...
Option 1) 13 : $1s^2 1p^6 2s^2 1d^3$ is not half filled
Option 2) 9 : $1s^2 1p^6 2s^1$ is the first alkali metal because after losing one electron, it will achieve first noble gas configuration
Option 3) 8 : $1s^2 1p^6$ is the first noble gas because after $1p^6$ e^- will enter 2s hence new period
Option 4) 6 : $1s^2 1p^4$ has 1p valence subshell.

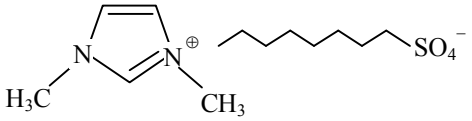
Q.5 An ionic micelle is formed on the addition of :

excess water to liquid

(1) 

excess water to liquid

(2) sodium stearate to pure toluene

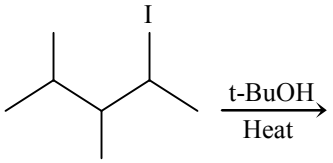
(3) 

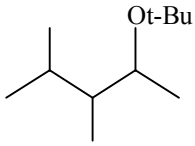
(4) liquid diethyl ether to aqueous NaCl solution

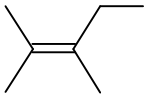
Ans. **[3]**

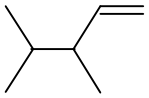
Sol. Correct Ans. is (3)

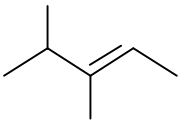
Q.6 The major product in the following reaction is :



(1) 

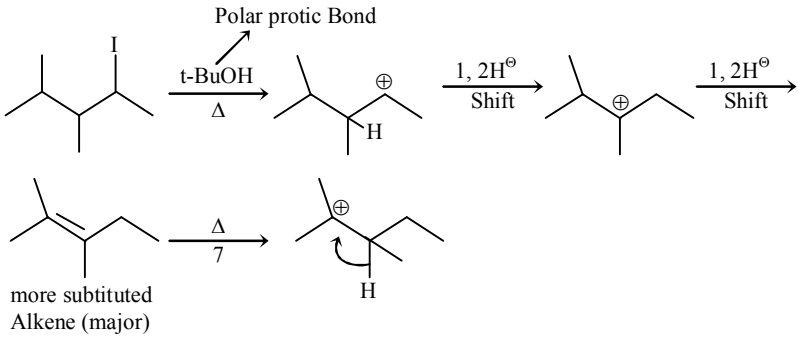
(2) 

(3) 

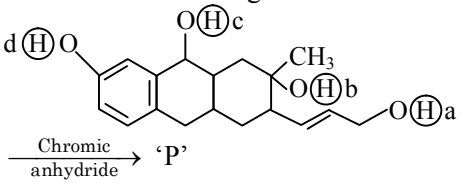
(4) 

Ans. **[2]**

Sol.



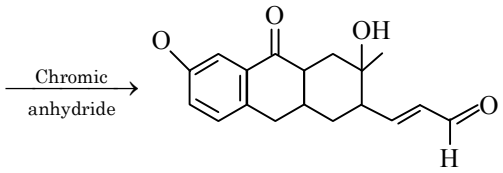
Q.7 Consider the following reaction :



The product 'P' gives positive ceric ammonium nitrate test. This is because of the presence of which of these –OH group(s) ?

(1) (b) only (2) (c) and (d) (3) (d) only (4) (b) and (d)

Ans. **[1]**

Sol. Compound 

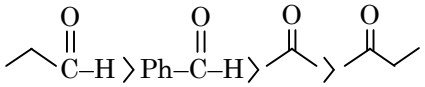
due to pressure of b.

Q.8 The increasing order of the reactivity of the following compounds in nucleophilic addition reaction is :
Propanal, Benzaldehyde, Propanone, Butanone

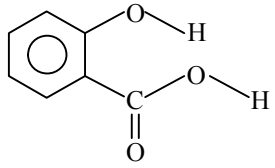
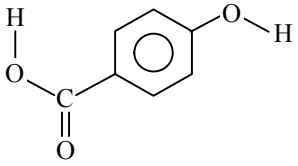
(1) Butanone < Propanone < Benzaldehyde < Propanal
(2) Propanal < Propanone < Butanone < Benzaldehyde
(3) Benaldehyde < Propanal < Propanone < Butanone
(4) Benzaldehyde < Butanone < Propanone < Propanal

Ans. **[1]**

Sol. Reactivity order of various carbonyl compounds → Aldehdes > Ketones



Q.9 Consider the following molecules and statements related to them :

(A)  (B) 

(a) (B) is more likely to be crystalline than (A)

(b) (B) has higher boiling point than (A)

(c) (B) dissolves more readily than (A) in water

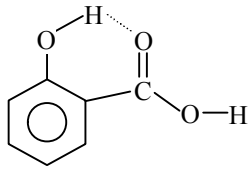
Identify the correct option from below :

(1) (a) and (b) are true (2) (a) and (c) are true

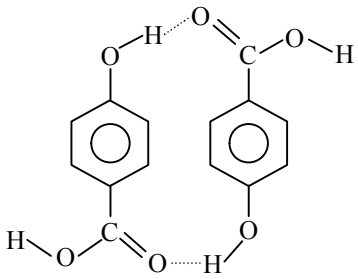
(3) Only (a) is true (4) (b) and (c) are true

Ans. **[4]**

Sol.



O-salicylic acid
intra molecular
H-bonding



p-salicylic acid
inter molecular
H-bonding

(a) B will be more crystalline due to more inter molecular interactions hence more efficient packing.

(b) B will have higher boiling point due to higher intermolecular interactions.

(c) B will be more soluble in water than A as B will have more extent of H-bonding in water

So all three statements are correct

{Solubility date $\Rightarrow$ O-salicylic acid = 2g/L ; P-salicylic acid = 5g/L}

Q.10 A mixture of one mole each of H_2, He andO_2 each are enclosed in a cylinder of volume V at temperature T. If the partial pressure of H_2 is 2 atm, the total pressure of the gases in the cylinder is -

(1) 14 atm (2) 22 atm (3) 38 atm (4) 6 atm

Ans. **[4]**

Sol. According to Dalton's law of partial pressure $p_i = x_i \times P_T$

p_i = partial pressure of the ith component

x_i = mole fraction of the ith component

p_T = total pressure of mixture

$$\Rightarrow \ 2 \text{ atm} = \left(\frac{n_{H_2}}{n_{H_2} + n_{H_e} + n_{O_2}} \right) \times p_T$$

$$\Rightarrow \ p_T = 2 \text{ atm} \times \frac{3}{1} = 6 \text{ atm}$$

Q.11 The d-electron configuration of $[Ru(en)_3]Cl_2$ and $[Fe(H_2O)_6]Cl_2$, respectively are :

(1) $t_{2g}^6\ e_g^0$ and $t_{2g}^4\ e_g^2$ (2) $t_{2g}^6\ e_g^0$ and $t_{2g}^6\ e_g^0$

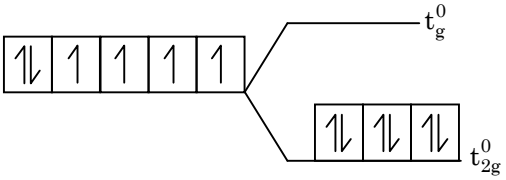
(3) $t_{2g}^4\ e_g^2$ and $t_{2g}^4\ e_g^2$ (4) $t_{2g}^4\ e_g^2$ and $t_{2g}^6\ e_g^0$

Ans. **[1]**

Sol. $[Ru(en)_3]\ Cl_2$ $R_u \Rightarrow$ 4d series

en $\Rightarrow$ chelating ligand

CN = 6, octahedral splitting hence laye splitting of d-subshell

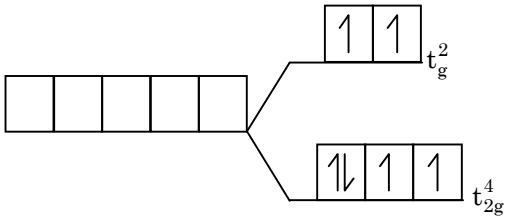
$R_u^{+2} \Rightarrow [Kr]4d^6 5s^0$ 

$[Fe(H_2O)_6]Cl_2 \Rightarrow H_2O \Rightarrow$ Weak filled ligand

$Fe^{+2} \Rightarrow [Ar]\ 3d^6 4s^0$

less plitting

CN = 6 octahedral splitting



Q.12 Match the following drugs with their therapeutic actions :

(i) Ranitidine	(a) Antidepressant
(ii) Nardil (Phenelzine)	(b) Antibioitic
(iii) Chloramphenicol	(c) Antihistamine
(iv) Dimetane (Brompheniramine)	(d) Antacid
	(e) Analgesic

(1) (i)-(e); (ii)-(a); (iii)-(c); (iv)-(d) (2) (i)-(d); (ii)-(c); (iii)-(a); (iv)-(e)

(3) (i)-(d); (ii)-(a); (iii)-(b); (iv)-(c) (4) (i)-(a); (ii)-(c); (iii)-(b); (iv)-(e)

Ans. **[3]**

Sol. Raniticline $\rightarrow$ Antacid

Nardl $\rightarrow$ Antidepressant

Chloramphenicol $\rightarrow$ Antibiotic

Dimetane $\rightarrow$ Antihistamine

Q.13 The incorrect statement is :

(1) Manganate and permanganate ions are paramagnetic

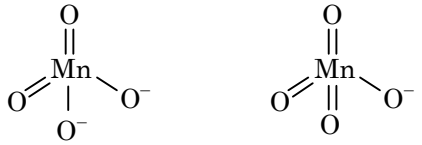
(2) Manganate ion is green in colour and permanganate ion is purple in colour

(3) In manganate and permanganate ions, the π-bonding, takes place by overtap of p-orbitals of oxygen and d-orbitals of manganese

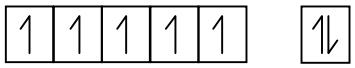
(4) Manganate and permanganate ions are tetrahedral

Ans. **[1]**

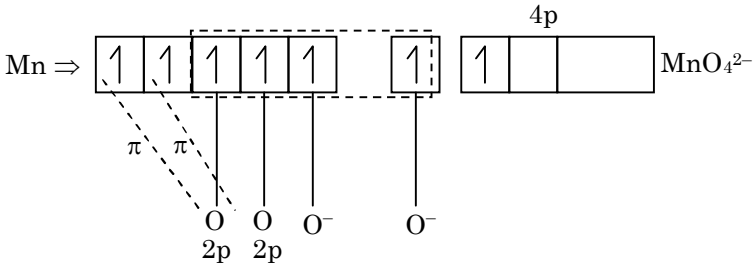
Sol. Option 1) Maganate ⇒ MnO_4^{2-},
Permanganate ⇒ MnO_4^-



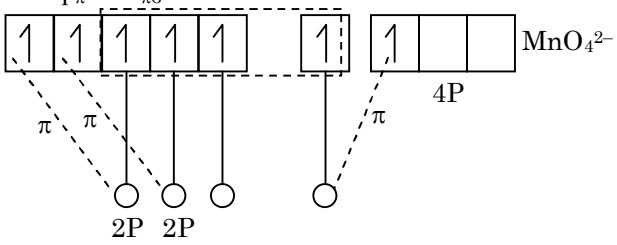
hybridisation of Mn ⇒ d^3s hybridisation of Mn ⇒ d^3s

Mn ⇒ 

After excitation



$2 \times 2p_\pi - 3d_{\pi\sigma}$



$2 \times 2P_\pi - 3d_\pi$
$1 \times 2P_\pi - 4P_\pi$

(2) MnO_4^{2-} ⇒ Green
MnO_4^- ⇒ purple/violet

(3) Manganate contains 1 unpaired electron hence it is paramagnetic.
where as permanganetic contains no unpaired electrons hence it is diamagnetic.

(4) Both have d^3s hybridisation hence both have tetrahedral geometry.

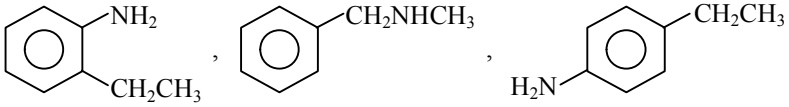
Q.14 Three isomers A, B and C (mol. formula $C_8H_{11}N$) give the following results :

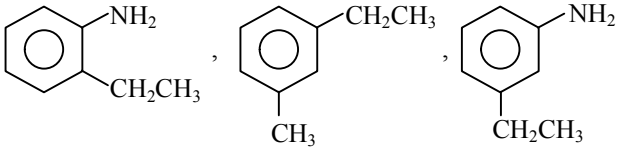
A and C $\xrightarrow{\text{Azotization}}$ P + Q $\xrightarrow[\substack{\text{(ii) Oxidation} \\ (kMnO_4+H^+)}]{\text{(i) Hydrolysis}}$ R(product of A) + S(product of C)

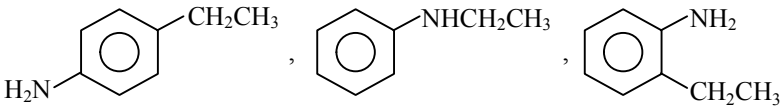
R has lower boiling point than S

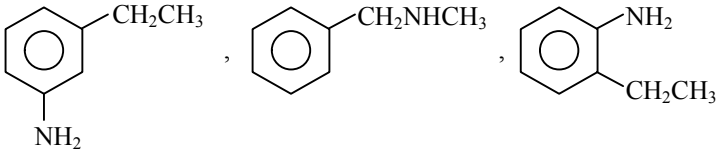
B $\xrightarrow{C_6H_5SO_2Cl}$ alkali-insoluble product

A, B and C, respectively are :

(1) 

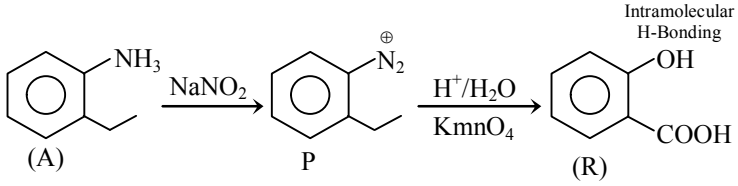
(2) 

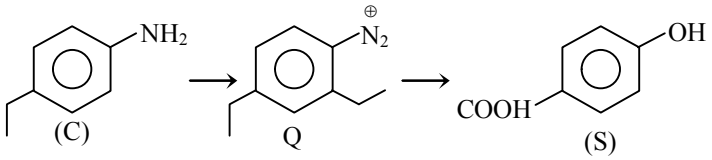
(3) 

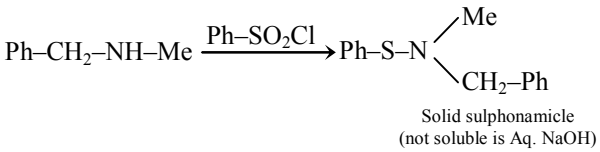
(4) 

Ans. **[1]**

Sol.







Q.15 Among the statements (I – IV), the correct ones are :

(I) Be has smaller atomic radius compared to Mg.

(II) Be has higher ionization enthalpy then Al.

(III) Charge/radius ratio of Be is greater than that of Al.

(IV) Both Be and Al form mainly covalent compounds.

(1) (I), (II) and (III) (2) (I), (II) and (IV)

(3) (I), (III) and (IV) (4) (II), (III) and (IV)

Ans. **[1]**

Sol. I, A_N ; Be < Mg

II IE : Be ,. Al

III Charge/radius ratio of Be w less than that of Al (IV) be, Al mainly form covalent compounds.

Q.16 The five successive ionization enthalpies of an element are 800, 2427, 3658, 25024 and 32824 kJ mol^{-1}. The number of valence electron in the element is -

(1) 3 (2) 4 (3) 5 (4) 2

Ans. **[1]**

Sol. Let suppose element X ⇒

$$X_{(g)} \xrightarrow[800]{IE_1} X(\overset{+}{g}) \xrightarrow[2427]{IE_2} X(\overset{+2}{g}) \xrightarrow[3658]{IE_3}$$

$$X(\overset{+3}{g}) \xrightarrow[25024]{IE_4} X(\overset{+4}{g}) \xrightarrow[32824]{IE_5} X(\overset{+5}{g})$$

X^{+3} has stable inert gas configuration as there is high jump after IE_3
So valence electrons are 3.

Q.17 Complex A has a composition of $H_{12}O_6Cl_3C_r$. If the complex on treatment with conc. H_2SO_4 loses 13.5 % of its original mass, the correct molecular formula of A is :
[Given : atomic mass of Cr = 52 amu and Cl = 35 amu]
(1) $[Cr(H_2O)_5Cl]\ Cl_2 \cdot H_2O$ (2) $[Cr(H_2O)_6]Cl_3$
(3) $[Cr(H_2O)_3Cl_3] \cdot 3H_2O$ (4) $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$

Ans. **[4]**

Sol. % mass of water

$$= \frac{x \times 18}{(12 + 6 \times + 35 \times 3 + 52)} \times 100 = 13.5$$

$$\Rightarrow x = \frac{265 \times 13.5}{18 \times 100} \simeq 2$$

Q.18 The strengths of 5.6 volume hydrogen peroxide (of density 1g/mL) in terms of mass percentage and molarity (M), respectively, are : (Take molar mass of hydrogen peroxide as 34 g/mol)
(1) 0.85 and 0.5 (2) 1.7 and 0.25 (3) 1.7 and 0.5 (4) 0.85 and 0.25

Ans. **[3]**

Sol. Volume strength = 11.2 × molarity

$$\Rightarrow \text{molarity} = \frac{5.6}{11.2} = 0.5$$

Assuming 1 litre solution;
mass of solution = 1000 ml × 1 g/ml = 1000 g
mass of solute = moles × molar mass
= 0.5 mol × 34 g/mol
= 17 gm.

$$\Rightarrow \text{mass\%} = \frac{17}{1000} \times 100 = 1.7\ \%$$

Q.19 For the reaction $2A + 3B + \frac{3}{2}C \rightarrow 3P$, which statement is correct?

(1) $\frac{dn_A}{dt} = \frac{2}{3}\frac{dn_B}{dt} = \frac{3}{4}\frac{dn_C}{dt}$ (2) $\frac{dn_A}{dt} = \frac{3}{2}\frac{dn_B}{dt} = \frac{3}{4}\frac{dn_C}{dt}$

(3) $\frac{dn_A}{dt} = \frac{dn_B}{dt} = \frac{dn_C}{dt}$ (4) $\frac{dn_A}{dt} = \frac{2}{3}\frac{dn_B}{dt} = \frac{4}{3}\frac{dn_C}{dt}$

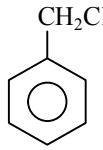
Ans. **[4]**

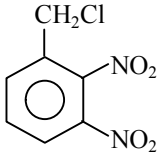
Sol. For $aA + bB \rightarrow cC$;

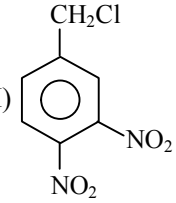
$$\frac{-1}{a}\frac{d[A]}{dt} = \frac{-1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt}$$

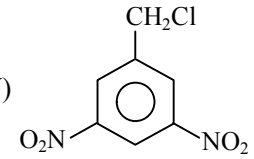
$$\therefore \frac{-1}{2}\frac{d[A]}{dt} = \frac{-1}{3}\frac{d[B]}{dt} = \frac{-2}{3}\frac{d[C]}{dt} = \frac{1}{3}\frac{d[p]}{dt}$$

Q.20 The decreasing order of reactivity of the following compounds towards nucleophilic substitution (S_N^2) is :

(I) 

(II) 

(III) 

(IV) 

(1) (IV) > (II) > (III) > (I)

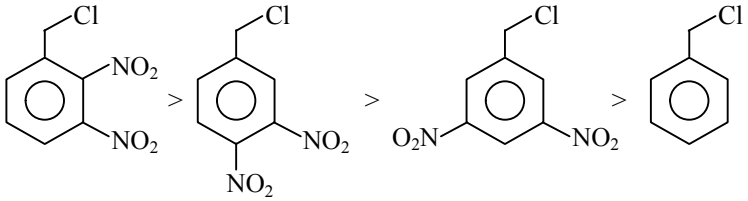
(2) (III) > (II) > (IV) > (I)

(3) (II) > (III) > (I) > (IV)

(4) (II) > (III) > (IV) > (I)

Ans. **[4]**

Sol.



–I & – R

less –I & –R
rate of SN^2 reaction

Only 2 – 1

Q.21 6.023×10^{22} molecules are present in 10 g of substance 'x'. The molarity of a solution containing 5 g of substance 'x' in 2L solution is .....$\times 10^{-3}$.

Ans. **[25]**

Sol. $$\text{moles} = \frac{\text{number of molecules}}{6\times10^{23}} = \frac{\text{given mass}}{\text{molar mass}}$$

$$\Rightarrow \text{molar mass} = \frac{10\times6.023\times10^{23}}{6.023\times10^{22}} = 100 \text{ g/mol}$$

$$\Rightarrow \text{molarity} = \frac{\text{moles of solute}}{\text{volume of sol}^n(\ell)} = \frac{(5/100)}{2} = 0.025$$

Q.22 An acidic solution of dichromate is electrolyzed for 8 minutes using 2A current. As per the following equation $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$. The amount of Cr^{3+} obtained was 0.104 g. The efficiency of the process (in %) is (Take : F = 96000 C, At. mass of chromium = 52) ..... .

Ans. **[60]**

Sol. $$\text{Moles of } e^{\Theta} = \left(\frac{8\times60\times2}{96000}\right)$$

Using stoichiometry; theoritically

$$\frac{n_{e^{\Theta}} \text{ used}}{6} = \frac{n_{cr^{+3}} \text{ produced}}{2}$$

$\Rightarrow \quad n_{cr^{+3}}$ produced $= \frac{2}{6} \times \frac{8 \times 60 \times 2}{96000}$

$\Rightarrow \quad wt_{cr^{+3}}$ theoritically produced $= \left(\frac{0.02}{6} \times 52\right) g$

$\Rightarrow$ % efficiency $= \frac{0.104g}{\left(\frac{0.02 \times 52}{6}\right)g} \times 100 = 60\ \%$

Q.23 The volume (in mL) of 0.1 N NaOH required to neutralise 10 mL of 0.1 N phosphinic acid is ..... .

Ans. **[10]**

Sol. $H_3PO_2 + NaOH \rightarrow NaH_2PO_2 + H_2O$

$$\frac{n_{H_3PO_2}\text{reached}}{1} = \frac{n_{NaOH}\text{reacted}}{1}$$

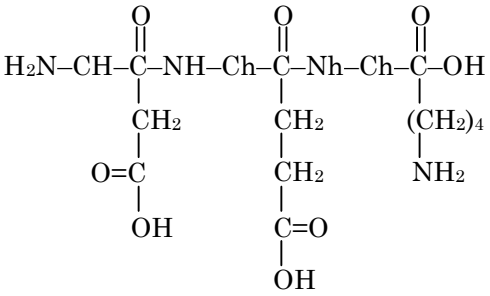
$$\Rightarrow \quad \frac{0.1 \times 10}{1} = 0.1 \times V_{NaOH}$$

$\Rightarrow \quad V_{NaOH} = 10$ ml.

Q.24 The number of C = O groups present in a tripeptide Aspp — Glu — Lys is ..... .

Ans. **[5]**

Sol. Structure of Tri peptide Asp – Glu – Lys



Q.25 If 250 cm^3 of an aqueous solution containing 0.73 g of a protein A is isotonic with one litre of another aqueous solution containing 1.66 g of a protein B. at 298 K, the ratio of the molecular masses of A and B is ..... $\times 10^{-2}$ (to the nearest integer).

Ans. **[177]**

Sol. Let molar mass of protein A = x g/mol

Let molar mass of protein B = y g/mol

π_A = osmotic pressure of protein A $= \frac{\left(\frac{0.73}{x}\right)}{0.25} RT$

π_B = osmotic pressure of protein B $= \frac{\left(\frac{1.65}{y}\right)}{1} RT$

$\pi_A = \pi_B$

$$\Rightarrow \quad \left(\frac{0.73}{x \times 0.25}\right) RT = \left(\frac{1.65}{y}\right) RT$$

$$\Rightarrow \quad \left(\frac{x}{y}\right) = \frac{0.73}{0.25 \times 1.65} = 1.769 \cong 1.77$$

CAREER POINT

JEE Main Online Exam 2020

Question with Solutions

3rd September 2020 | Shift-II

MATHEMATICS

Q.1 Let e_1 and e_2 be the eccentricities of the ellipse, $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ $(b < 5)$ and the hyperbola, $\frac{x^2}{16} - \frac{y^2}{b^2} = 1$ respectively satisfying $e_1e_2 = 1$. If α and β are the distances between the foci of the ellipse and the foci of the hyperbola respectively, then the ordered pair (α, β) is equal to -

(1) $\left(\frac{24}{5}, 10\right)$ (2) (8, 12) (3) $\left(\frac{20}{3}, 12\right)$ (4) (8, 10)

Ans. **[4]**

Sol. For ellipse $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ $(b < 5)$

Let e_1 is eccentricity of ellipse

$\therefore$ $b^2 = 25\,(1 - e_1^2)$ ....(1)

Again for hyperbola

$$\frac{x^2}{16} - \frac{y^2}{b^2} = 1$$

Let e_2 is eccentricity of hyperbola.

$\therefore$ $b^2 = 16(e_2^2 - 1)$ ....(2)

by (1) & (2)

$25\,(1 - e_1^2) = 16(e_2^2 - 1)$

Now $e_1 \cdot e_2 = 1$ (given)

$\therefore$ $25(1 - e_1^2) = 16\left(\frac{1 - e_1^2}{e_1^2}\right)$

or $e_1 = \frac{4}{5}$ $\therefore$ $e_2 = \frac{5}{4}$

Now distance between foci is 2ae

$\therefore$ distance for ellipse $= 2 \times 5 \times \frac{4}{5} = 8 = \alpha$

distance for hyperbola $= 2 \times 4 \times \frac{5}{4} = 10 = \beta$

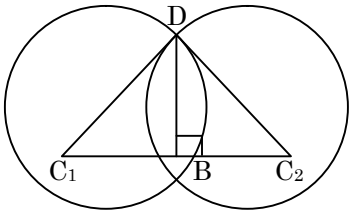
$\therefore$ $(\alpha, \beta) \equiv (8, 10)$

Q.2 Let the latus rectum of the parabola $y^2 = 4x$ be the common chord to the circles C_1 and C_2 each of them having radius $2\sqrt{5}$. Then, the distance between the centres of the circles C_1 and C_2 is -

(1) $4\sqrt{5}$ (2) 8 (3) $8\sqrt{5}$ (4) 12

Ans. **[2]**

Sol. Length of latus rectum = 4



DB = 2

$C_1B = \sqrt{(C_1D)^2 - (DB)^2} = 4$

$C_1C_2 = 8$

Q.3 If the value of the integral $\int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}}dx$ is $\frac{k}{6}$, then k is equal to -

(1) $2\sqrt{3} + \pi$ (2) $3\sqrt{2} - \pi$ (3) $3\sqrt{2} + \pi$ (4) $2\sqrt{3} - \pi$

Ans. **[4]**

Sol. $\int_0^{1/2} \frac{((x^2-1)+1)}{(1-x^2)^{3/2}}dx$

$\int_0^{1/2} \frac{dx}{(1-x^2)^{3/2}} - \int_0^{1/2} \frac{dx}{\sqrt{1-x^2}}$

$\int_0^{1/2} \frac{x^{-3}}{(x^{-2}-1)^{3/2}}\,dx - (\sin^{-1}x)_0^{1/2}$

Let $x^{-2} - 1 = t^2 \Rightarrow x^{-3}\,dx = -t\,dt$

$\int_\infty^{\sqrt{3}} \frac{-t\,dt}{t^3} - \frac{\pi}{6} = \int_{\sqrt{3}}^{\infty} \frac{dt}{t^2} - \frac{\pi}{6} = \frac{1}{\sqrt{3}} - \frac{\pi}{6} = \frac{k}{6}$

$k = 2\sqrt{3} - \pi$

Q.4 Let x_i $(1 \le i \le 10)$ be ten observations of a random variable X. If $\sum_{i=1}^{10}(x_i - p) = 3$ and $\sum_{i=1}^{10}(x_i - p)^2 = 9$ where $0 \ne p \in R$, then the standard deviation of these observations is :

(1) $\sqrt{\frac{3}{5}}$ (2) $\frac{9}{10}$ (3) $\frac{4}{5}$ (4) $\frac{7}{10}$

Ans. **[2]**

Sol. Variance $= \frac{\Sigma(x_i - p)^2}{n} - \left(\frac{\Sigma(x_i - p)}{n}\right)^2$

$= \frac{9}{10} - \left(\frac{3}{10}\right)^2 = \frac{81}{100}$

S.D. $= \frac{9}{10}$

Q.5 Let R_1 and R_2 be two relations defined as follows :
$R_1 = [(a, b) \in R^2 ; a^2 + b^2 \in Q]$ and $R_2 = [(a, b) \in R^2 : a^2 + b^2 \notin Q]$, where Q is the set of all rational numbers. Then :

(1) Neither R_1 nor R_2 is transitive.
(2) R_1 and R_2 are both transitive
(3) R_1 is transitive but R_2 is not transitive
(4) R_2 is transitive but R_1 is not transitive

Ans. **[1]**

Sol. Let $a^2 + b^2 \in Q$ & $b^2 + c^2 \in Q$

eg. $a = 2 + \sqrt{3}$ & $b = 2 - \sqrt{3}$

$a^2 + b^2 = 14 \in Q$

Let $c = (1 + 2\sqrt{3})$

$b^2 + c^2 = 20 \in Q$

But $a^2 + c^2 = (2 + \sqrt{3})^2 + (1 + 2\sqrt{3})^2 \notin Q$

for R_2 Let $a^2 = 1$, $b^2 = \sqrt{3}$ & $c^2 = 2$

$a^2 + b^2 \notin Q$ & $b^2 + c^2 \notin Q$

But $a^2 + c^2 \in Q$

Q.6 The probability that a randomly chosen 5-digit number is made from exactly two digits is :

(1) $\frac{121}{10^4}$ (2) $\frac{134}{10^4}$ (3) $\frac{150}{10^4}$ (4) $\frac{135}{10^4}$

Ans. **[4]**

Sol. First Case : Choose two non-zero digits ${}^9C_2$.

Now, number of 5-digit numbers containing both digits $= 2^5 - 2$

Second Case : Choose one non-zero & one zero as digit ${}^9C_1$.

Number of 5-digit numbers containing one non zero and one zero both $= (2^4 - 1)$

Required prob.

$= \frac{({}^9C_2 \times (2^5 - 2) + {}^9C_1 \times (2^4 - 1))}{9 \times 10^4}$

$= \frac{36 \times (32 - 2) + 9 \times (16 - 1)}{9 \times 10^4}$

$= \frac{4 \times 30 + 15}{10^4} = \frac{135}{10^4}$

Q.7 Let p, q, r be three statements such that the truth value of $(p \wedge q) \rightarrow (\sim q \vee r)$ is F. Then the truth values of p, q, r are respectively.

(1) T, F, T (2) T, T, T (3) T, T, F (4) F, T, F

Ans. **[3]**

Sol. $(p \wedge q) \rightarrow (\sim q \vee r)$ = false

when $(p \wedge q) = T$

and $(\sim q \vee r) = F$

So $(p \wedge q) = T$ is possible when p = q = true

$\therefore$ $\sim q$ = False (q = true)

So $(\sim q \vee r)$ = False is possible if r is false

$\therefore$ p = T, q = T, r = F

Q.8 Let a, b, c $\in$ R be such that $a^2 + b^2 + c^2 = 1$. If $a\cos\theta = b\cos\left(\theta + \frac{2\pi}{3}\right) = c\cos\left(\theta + \frac{4\pi}{3}\right)$, where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ is -

(1) $\frac{\pi}{2}$ (2) $\frac{2\pi}{3}$ (3) $\frac{\pi}{9}$ (4) 0

Ans. **[1]**

Sol. $\cos\phi = \frac{\bar{p}.\bar{q}}{|\bar{p}||\bar{q}|} = \frac{ab + bc + ca}{a^2 + b^2 + c^2} = \frac{\Sigma ab}{1}$

$= abc\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$

$= \frac{abc}{\lambda}\left(\cos\theta + \cos\left(\theta + \frac{2\pi}{3}\right) + \cos\left(\theta + \frac{4\pi}{3}\right)\right)$

$= \frac{abc}{\lambda}\left(\cos + 2\cos(\theta + \pi)\cos\frac{\pi}{3}\right)$

$= \frac{abc}{\lambda}(\cos\theta - \cos\theta) = 0$

$\phi = \frac{\pi}{2}$

Q.9 If the sum of the series $20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} + \ldots$ upto n^{th} term is 488 and the nth term is negative, then-

(1) n^{th} term is $-4\frac{2}{5}$ (2) n = 41 (3) n^{th} term is –4 (4) n = 60

Ans. **[3]**

Sol. $S = \frac{100}{5} + \frac{98}{5} + \frac{96}{5} + \frac{94}{5} + \ldots. n$

$S_n = \frac{n}{2}\left(2\times\frac{100}{5} + (n-1)\left(\frac{-2}{5}\right)\right) = 188$

$n(100 - n + 1) = 488 \times 5$

$n^2 - 101n + 488 \times 5 = 0$

n = 61, 40

$T_n = a + (n-1)d = \frac{100}{5} - \frac{2}{5} \times 60$

$= 20 - 24 = -4$

Q.10 If z_1, z_2 are complex numbers such that $\text{Re}(z_1) = |z_1 - 1|$, $\text{Re}(z_2) = |z_2 - 1|$ and $\arg(z_1 - z_2) = \frac{\pi}{6}$, then $\text{Im}(z_1 + z_2)$ is equal to -

(1) $\frac{2}{\sqrt{3}}$ (2) $2\sqrt{3}$ (3) $\frac{1}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2}$

Ans. **[2]**

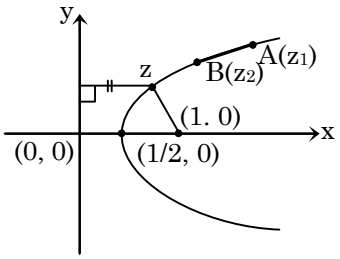
Sol. $\text{Re}(z) = |z - 1|$

$\Rightarrow x = \sqrt{(x-1)^2 + (y-0)^2}$ $(x > 0)$

$\Rightarrow y^2 = 2x - 1 = 4\,\frac{1}{2}\left(x - \frac{1}{2}\right)$

$\Rightarrow$ a parabola with focus (1, 0) & directrix as imaginary axis.

$\therefore$ Vertex $= \left(\frac{1}{2}, 0\right)$



$A(z_1)$ & $B(z_2)$ are two points on it such that slope of AB $= \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$

$(\arg(z_1 - z_2) = \frac{\pi}{6})$

for $y^2 = 4ax$

Let $A(at_1^2, 2at_1)$ & $B(at_2^2, 2at_2)$

$m_{AB} = \frac{2}{t_1 + t_2} = \frac{4a}{y_1 + y_2} = \frac{1}{\sqrt{3}}$

$\left(\text{Here } a = \frac{1}{2}\right)$

$\Rightarrow y_1 + y_2 = 4a\sqrt{3} = 2\sqrt{3}$

Q.11 If $\int \sin^{-1}\left(\sqrt{\frac{x}{1+x}}\right)dx = A(x)\tan^{-1}(\sqrt{x}) + B(x) + C$, where C is a constant of integration, then the ordered pair (A(x), B(x)) can be -

(1) $(x + 1, \sqrt{x})$ (2) $(x + 1, -\sqrt{x})$ (3) $(x - 1, -\sqrt{x})$ (4) $(x - 1, \sqrt{x})$

Ans. **[2]**

Sol. Put $x = \tan^2\theta \Rightarrow dx = 2\tan\theta\sec^2\theta\, d\theta$

$$\int \theta.(2\tan\theta.\sec^2\theta)\,d\theta$$

$\downarrow$ I $\quad \downarrow$ II (By parts)

$$= \theta \cdot \tan^2\theta - \int \tan^2\theta\, d\theta$$

$$= \theta \cdot \tan^2\theta - \int (\sec^2\theta - 1)\,d\theta$$

$$= \theta(1 + \tan^2\theta) - \tan\theta + C$$

$$= \tan^{-1}(\sqrt{x})\,(1 + x) - \sqrt{x} + C$$

Q.12 $\lim\limits_{x\to a} \dfrac{(a+2x)^{1/3} - (3x)^{1/3}}{(3a+x)^{1/3} - (4x)^{1/3}}$ $(a \neq 0)$ is equal to -

(1) $\left(\frac{2}{3}\right)\left(\frac{2}{9}\right)^{1/3}$ (2) $\left(\frac{2}{9}\right)^{4/3}$ (3) $\left(\frac{2}{9}\right)\left(\frac{2}{3}\right)^{1/3}$ (4) $\left(\frac{2}{3}\right)^{4/3}$

Ans. **[1]**

Sol. Required limit

$$L = \lim_{h\to 0} \frac{(a+2(a+h))^{1/3} - (3(a+h))^{1/3}}{(3a+a+h)^{1/3} - (4(a+h))^{1/3}}$$

$$= \lim_{h\to 0} \frac{(3a)^{1/3}\left(1+\frac{2h}{3a}\right)^{1/3} - (3a)^{1/3}\left(1+\frac{h}{a}\right)^{1/3}}{(4a)^{1/3}\left(1+\frac{h}{4a}\right)^{1/3} - (4a)^{1/3}\left(1+\frac{h}{a}\right)^{1/3}}$$

$$= \lim_{h\to 0} \left(\frac{3^{1/3}}{4^{1/3}}\right)\left[\frac{\left(1+\frac{2h}{9a}\right) - \left(1+\frac{h}{3a}\right)}{\left(1+\frac{h}{12a}\right) - \left(1+\frac{h}{3a}\right)}\right]$$

$$= \left(\frac{3}{4}\right)^{1/3} \frac{\left(\frac{2}{9} - \frac{1}{3}\right)}{\left(\frac{1}{12} - \frac{1}{3}\right)} = \left(\frac{3}{4}\right)^{1/3} \left(\frac{8-12}{3-12}\right)$$

$$= \left(\frac{3}{4}\right)^{1/3} \left(\frac{-4}{-9}\right) = \frac{4^{1-\frac{1}{3}}}{3^{2-\frac{1}{3}}} = \frac{4^{2/3}}{3^{5/3}}$$

$$= \frac{(8\times 2)^{1/3}}{(27\times 9)^{1/3}} = \frac{2}{3}\left(\frac{2}{9}\right)^{1/3}$$

Q.13 If the term independent of x in the expansion of $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is k, then 18 k is equal to -

(1) 5 (2) 11 (3) 7 (4) 9

Ans. **[3]**

Sol. $T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(-\frac{1}{3x}\right)^r$

$T_{r+1} = {}^9C_r \left(\frac{3}{2}\right)^{9-r} \left(-\frac{1}{3}\right)^r x^{18-3r}$

For independent of x

$18 - 3r = 0,\ r = 6$

$\therefore \quad T_7 = {}^9C_6 \left(\frac{3}{2}\right)^3 \left(-\frac{1}{3}\right)^6 = \frac{21}{54} = k$

$\therefore \quad 18k = \frac{21}{54} \times 18 = 7$

Q.14 If $x^3dy + xy\,dx = x^2dy + 2y\,dx$; $y(2) = e$ and $x > 1$, then $y(4)$ is equal to -

(1) $\frac{\sqrt{e}}{2}$ (2) $\frac{3}{2}\sqrt{e}$ (3) $\frac{3}{2} + \sqrt{e}$ (4) $\frac{1}{2} + \sqrt{e}$

Ans. **[2]**

Sol. $x^3dy + xy\,dx = x^2\,dy + 2y\,dx$

$\Rightarrow \quad dy(x^3 - x^2) = dx\,(2y - xy)$

$\Rightarrow \quad -\int\frac{1}{y}dx = \int\frac{x-2}{x^2(x-1)}dx$

$\Rightarrow \quad -\ell ny = \int\left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{(x-1)}\right)dx$

Where $A = 1,\ B = +2,\ C = -1$

$\Rightarrow \quad -\ell ny = \ell nx - \frac{2}{x} - \ell n\,(x-1) + \lambda$

$\Rightarrow \quad y(2) = e$

$\Rightarrow \quad -1 = \ell n\,2 - 1 - 0 + \lambda$

$\therefore \quad \lambda = -\ell n\,2$

$\Rightarrow \quad \ell ny = -\ell nx + \frac{2}{x} + \ell n\,(x-1) + \ell n\,2$

Now put $x = 4$ in equation

$\Rightarrow \quad \ell ny = -\ell n4 + \frac{1}{2} + \ell n\,3 + \ell n\,2$

$\Rightarrow \quad \ell ny = \ell n\left(\frac{3}{2}\right) + \frac{1}{2}\,\ell ne$

$\Rightarrow \quad y = \frac{3}{2}\sqrt{e}$

Q.15 Let A be a 3 × 3 matrix such that adj A = $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ and B = adj (adj A). If $|A| = \lambda$ and $|(B^{-1})^T| = \mu$, then the ordered pair, $(|\lambda|, \mu)$ is equal to :

(1) $\left(9, \frac{1}{81}\right)$ (2) (3, 81) (3) $\left(9, \frac{1}{9}\right)$ (4) $\left(3, \frac{1}{81}\right)$

Ans. **[4]**

Sol. C = adj. A = $\begin{vmatrix} +2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{vmatrix}$

$|C| = |\text{adj } A| = +2(0 + 4) + 1 \,.\, (1 - 2) + 1.\, (2, 4)$

$= + 8 - 1 + 2$

$|\text{adj } A| = |A|^2 = 9 = 9$

$\lambda = |A| = \pm 3$

$|\lambda| = 3$

B = adj C

$|B| = |\text{adj } C| = |C|^2 = 81$

$|(B^{-1})^T| = |B|^{-1} = \frac{1}{81}$

$(|\lambda|, \mu) = \left(3, \frac{1}{81}\right)$

Q.16 If the surface area of a cube is increasing at a rate of 3.6 cm^2/sec, retaining its shape, then the rate of change of its volume (in cm^3/sec.), when the length of a side of the cube is 10 cm, is :

(1) 18 (2) 20 (3) 10 (4) 9

Ans. **[4]**

Sol. $\frac{d}{dt}(6a^2) = 3.6 \Rightarrow 12a \frac{da}{dt} = 3.6$

$a \frac{da}{dt} = 0.3$

$\frac{dv}{dt} = \frac{d}{dt}(a^3) = 3a\left(a\frac{da}{dt}\right)$

$= 3 \times 10 \times 0.3 = 9$

Q.17 Suppose f(x) is a polynomial of degree four, having critical points at –1, 0, 1. If $T = \{x \in R | f(x) = f(0)\}$, then the sum of squares of all the elements of T is :

(1) 8 (2) 2 (3) 6 (4) 4

Ans. **[4]**

Sol. $f'(x) = x(x+1)(x-1) = x^3 - x$

$$\int df(x) = \int x^3 - x\,dx$$

$$f(x) = \frac{x^4}{4} - \frac{x^2}{2} + C$$

$f(x) = f(0)$

$$\frac{x^4}{4} - \frac{x^2}{2} = 0$$

$x^2(x^2 - 2) = 0$

$x = 0, 0, \sqrt{2}, -\sqrt{2}$

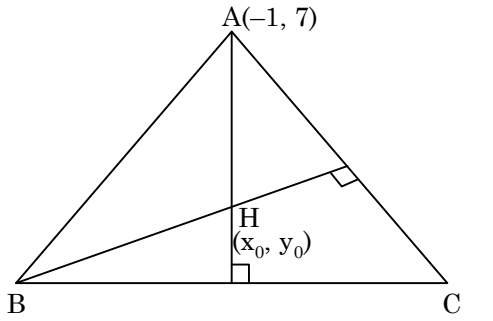
$x_1^2 + x_2^2 + x_3^2 = 0 + 2 + 2 = 4$

Q.18 If a ΔABC has vertices A(–1, 7), B(–7,1) and C(5, –5), then its orthocentre has coordinates -

(1) $\left(\frac{3}{5}, -\frac{3}{5}\right)$ (2) (3, –3) (3) $\left(-\frac{3}{5}, \frac{3}{5}\right)$ (4) (–3, 3)

Ans. **[4]**

Sol. Let orthocentre is $H(x_0, y_0)$



$m_{AH} \cdot m_{BC} = -1$

$$\Rightarrow \left(\frac{y_0 - 7}{x_0 + 1}\right)\left(\frac{1+5}{-7-5}\right) = -1$$

$\Rightarrow 2x_0 - y_0 + 9 = 0$ ....(1)

and $m_{BH} \cdot m_{AC} = -1$

$$\Rightarrow \left(\frac{y_0 - 1}{x_0 + 7}\right)\left(\frac{7-(-5)}{-1-5}\right) = -1$$

$\Rightarrow x_0 - 2y_0 + 9 = 0$ ....(2)

Solving equation (1) and (2) we get

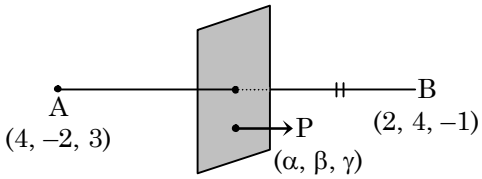
$(x_0, y_0) \equiv (-3, 3)$

Q.19 The plane which bisects the line joining the points (4, –2, 3) and (2, 4, –1) at right angles also passes through the point -

(1) (4, 0, –1) (2) (4, 0, 1) (3) (0, 1, –1) (4) (0, –1, 1)

Ans. **[1]**

Sol.



$PA = PB$

$\Rightarrow PA^2 = PB^2$

$\Rightarrow (\alpha - 4)^2 + (\beta + 2)^2 + (\gamma + 3)^2$

$= (\alpha - 2)^2 + (\beta - 4)^2 + (\gamma + 1)^2$

$\Rightarrow -4\alpha + 12\beta - 8\gamma = -8$

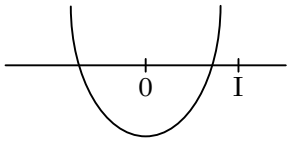
$\Rightarrow 2x - 6y + 4z = 4$

Q.20 The set of all real values of λ for which the quadratic equations, $(\lambda^2 + 1)\, x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval (0, 1) is :

(1) (–3, –1) (2) (2, 4] (3) (1, 3] (4) (0, 2)

Ans. **[3]**

Sol. If exactly one root in (0, 1) then



$\Rightarrow f(0) \cdot f(1) < 0$

$\Rightarrow 2(\lambda^2 - 4\lambda + 3) < 0$

$\Rightarrow 1 < \lambda < 3$

Now for $\lambda = 1$, $2x^2 - 4x + 2 = 0$

$(x - 1)^2 = 0$, $x = 1, 1$

So both roots doesn't lie between (0, 1)

$\therefore \lambda \neq 1$

Again for $\lambda = 3$

$10x^2 - 12x + 2 = 0$

$\Rightarrow x = 1, \frac{1}{5}$

So if one root is 1 then second root lie between (0, 1) so $\lambda = 3$ is correct.

$\therefore \lambda \in (1, 3]$.

Q.21 If the tangent to the curve, $y = e^x$ at a point (c, e^c) and the normal to the parabola, $y^2 = 4x$ at the point (1, 2) intersect at the same point on the x-axis, then the value of c is ..... .

Ans. **[4]**

Sol. $y = e^x \Rightarrow \frac{dy}{dx} = e^x$

$m = \left(\frac{dy}{dx}\right)_{(c, e^c)} = e^c$

$\Rightarrow$ Tangent at (c, e^c)

$y - e^c = e^c (x - c)$

it intersect x-axis

Put $y = 0 \Rightarrow x = c - 1$ ....(1)

Now $y^2 = 4x \Rightarrow \frac{dy}{dx} = \frac{2}{y} \Rightarrow \left(\frac{dy}{dx}\right)_{(1, 2)} = 1$

$\Rightarrow$ Slope of normal = –1

Equation of normal $y - 2 = -1(x - 1)$

$x + y = 3$ it intersect x-axis

Put $y = 0 \Rightarrow x = 3$ ....(2)

Points are same

$\Rightarrow x = c - 1 = 3$

$\Rightarrow c = 4$

Q.22 If m arithmetic means (A. Ms) and three geometric means (G. Ms) are inserted between 3 and 243 such that 4th A.M. is equal to 2nd G.M., then m is equal to ..... .

Ans. **[39]**

Sol. $3, A_1, A_2 \ldots\ldots A_m, 243$

$d = \frac{243 - 3}{m + 1} = \frac{240}{m + 1}$

Now $3, G_1, G_2, G_3, 243$

$r = \left(\frac{243}{3}\right)^{\frac{1}{3+1}} = 3$

$\therefore A_4 = G_2$

$\Rightarrow a + 4d = at^2$

$3 + 4\left(\frac{240}{m + 1}\right) = 3\,(3)^2$

$m = 39$

Q.23 The total number of 3-digit numbers, whose sum of digits is 10, is ..... .

Ans. **[54]**

Sol. Let three digit number is xyz

$x + y + z = 10;\ x \geq 1,\ y \geq 0\ z \geq 0$ ....(1)

Let $T = x - 1 \Rightarrow x = T + 1$ where $T \geq 0$

Put in (1)

$T + y + z = 9;\ 0 \leq T \leq 8,\ 0 \leq y,\ z \leq 9$

No. of non negative integral solution

$= {}^{9+3-1}C_{3-1} - 1$ (when T = 9)

$= 55 - 1 = 54$

Q.24 Let S be the set of all integer solutions, (x, y, z), of the system of equations

$x - 2y + 5z = 0$

$-2x + 4y + z = 0$

$-7x + 14y + 9z = 0$

such that $15 \le x^2 + y^2 + z^2 \le 150$. Then, the number of elements in the set S is equal to ..... .

Ans. **[8]**

Sol. $\Delta = \begin{vmatrix} 1 & -2 & 5 \\ -2 & 4 & 1 \\ -7 & 14 & 9 \end{vmatrix} = 0$

Let $x = k$

$\Rightarrow$ Put in (1) & (2)

$k - 2y + 5z = 0$

$-2k + 4y + z = 0$

$z = 0, y = \frac{k}{2}$

$\therefore$ x, y, z are integer

$\Rightarrow$ k is even integer

Now $x = k, y = \frac{k}{2}, z = 0$ put in condition

$$15 \le k^2 + \left(\frac{k}{2}\right)^2 + 0 \le 150$$

$$12 \le k^2 \le 120$$

$\Rightarrow$ $k = \pm 4, \pm 6, \pm 8, \pm 10$

$\Rightarrow$ Number of element in S = 8.

Q.25 Let a plane P contain two lines $\vec{r} = \hat{i} + \lambda(\hat{i} + \hat{j}), \lambda \in R$ and $\vec{r} = -\hat{j} + \mu(\hat{j} - \hat{k}), \mu \in R$. If Q (α, β, γ) is the foot of the perpendicular drawn from the point M(1, 0, 1) to P, then $3(\alpha + \beta + \gamma)$ equals ..... .

Ans. **[5]**

Sol. Dr's normal to plane

$$= \begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} = -\hat{i} + \hat{j} + \hat{k}$$

Equation of plane

$-1(x - 1) + 1(y - 0) + 1(z - 0) = 0$

$x - y - z - 1 = 0$ ....(1)

Now $\frac{\alpha - 1}{1} = \frac{\beta - 0}{-1} = \frac{\gamma - 1}{-1} = \frac{(1 - 0 - 1 - 1)}{3}$

$\frac{\alpha - 1}{1} = \frac{\beta}{-1} = \frac{\gamma - 1}{-1} = \frac{1}{3}$

$\alpha = \frac{4}{3}, \beta = -\frac{1}{3}, \gamma = \frac{2}{3}$

$3(\alpha + \beta + \gamma) = 3\left(\frac{4}{3} - \frac{1}{3} + \frac{2}{3}\right) = 5$